Section: Module 6

Regression Discontinuity Design

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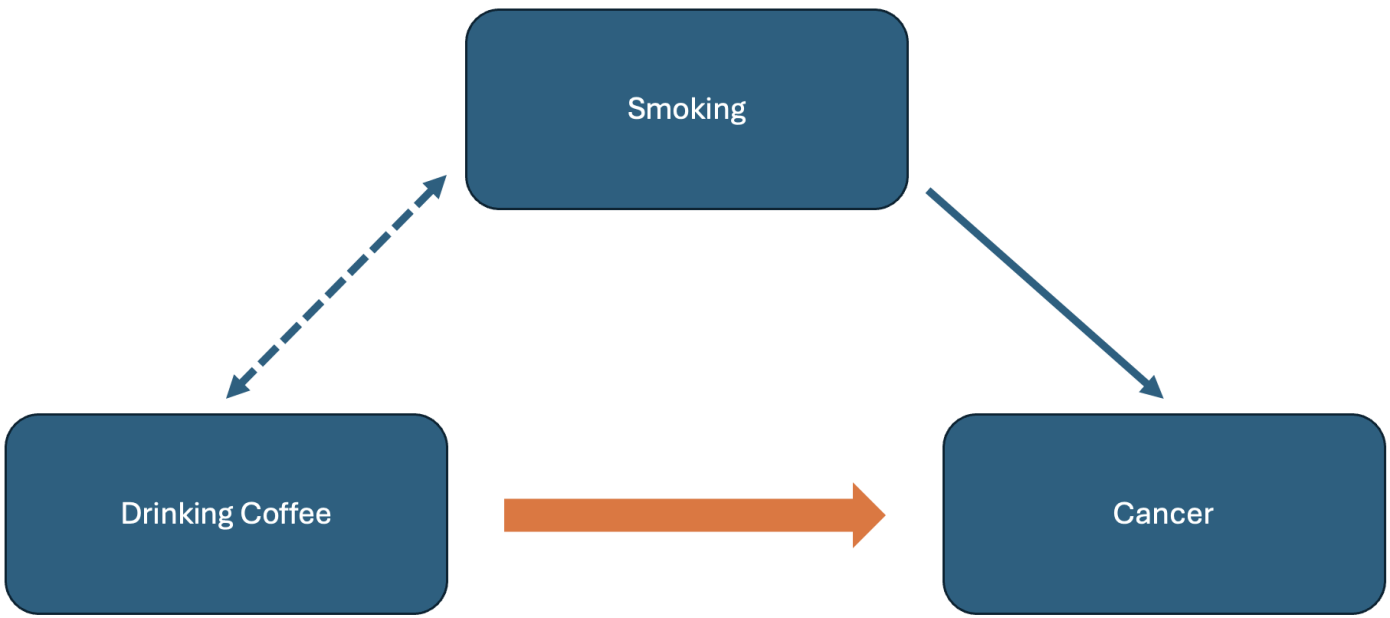
Today's Agenda

- Overview of Observational Studies
- Sharp RDD
  - Basic Setup / Intuition
  - Estimand / Assumption
  - Identification
  - Estimation
- Fuzzy RDD
- Review Question

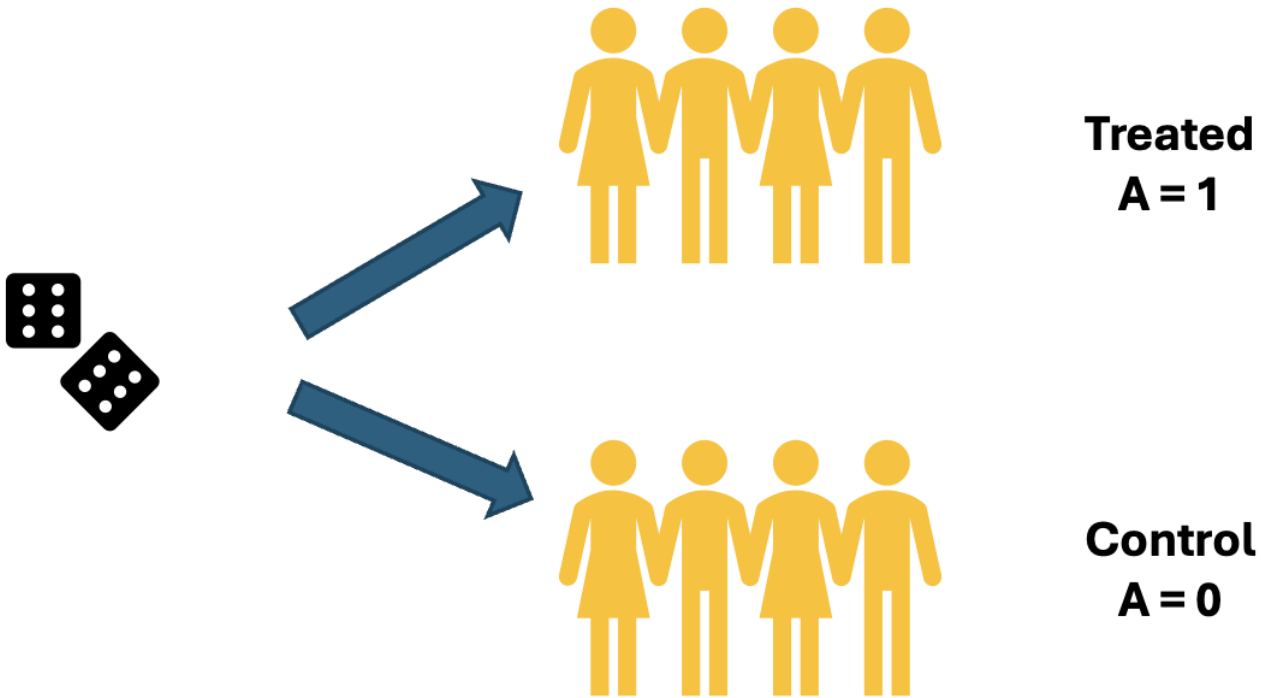
Observational Studies: Overview

- From this week, we are in observational studies
  - Difference: Lack of randomization

- Example: Does drinking coffee cause cancer?



Observational Studies: Overview

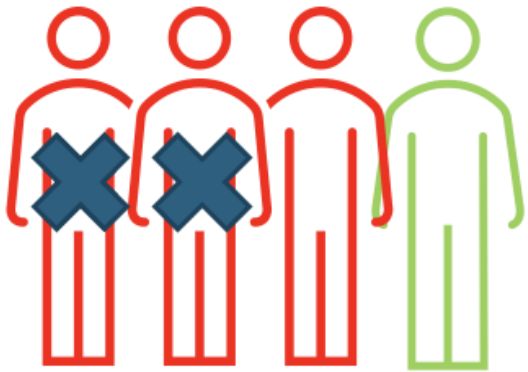


- Randomization makes two group comparable
  - Thus, difference-in-means works!

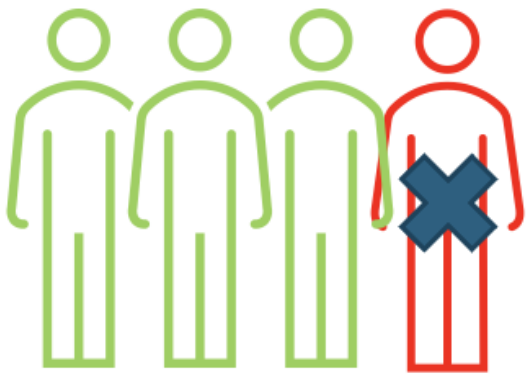
Observational Studies: Overview

- It is not the case for observational studies

Drinking Coffee
A = 1

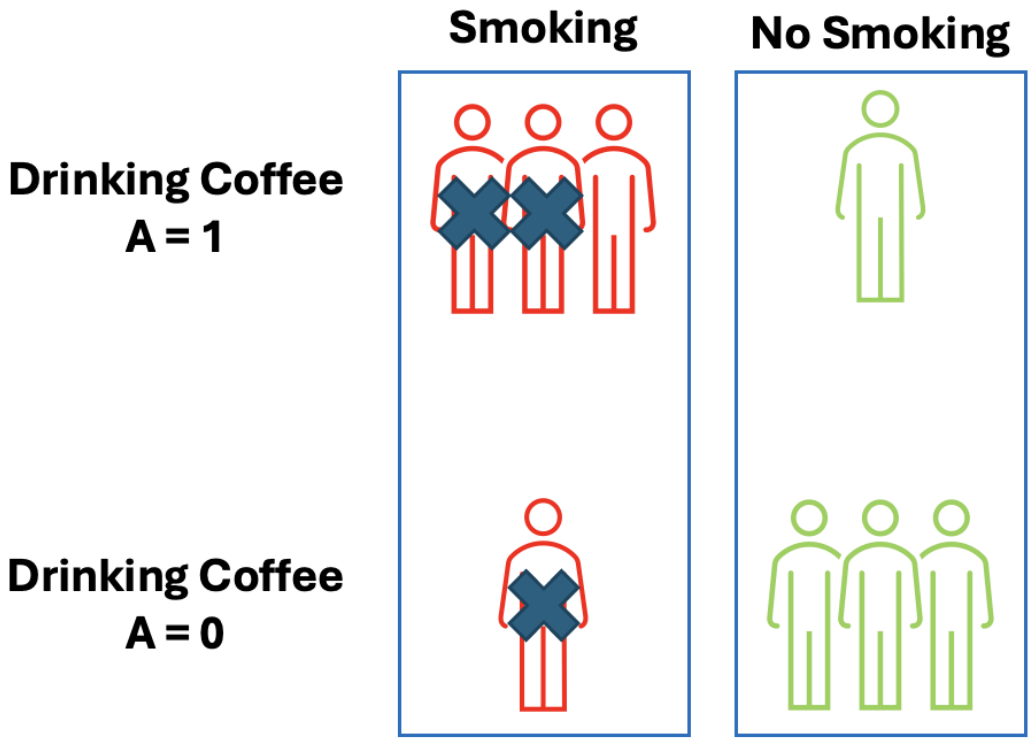


Drinking Coffee
A = 0



Observational Studies: Overview

- Basic Approach: Control all the confounders
  - **Confounders**: Variables affecting both treatment and outcome



- Requirement: Observe all the confounders
  - Also, within each strata, there must exist counterparts of the comparison

Observational Studies: Overview

- For confounders C_i, we formally need the followings:

$$\text{Ignorability / Exchangeability}: \quad Y_i(t) \perp\!\!\!\perp T_i \mid C_i$$
$$\text{Positivity / Common Support}: \quad 0 < \mathbb{P}(T_i = 1 \mid C_i) < 1$$

- We will cover these strategies in future modules
  - Module 8: Controlling confounders
    - Different ways of modeling (e.g., matching, weighting, regression)
  - Module 7: Sensitivity analysis + Partial identification
    - Checking credibility of assumptions
- BUT this is really hard! We rarely observe all the confounders
- **Quasi-experimental design**: Since we have no randomization, we will use several designs to estimate the causal effect
  - Module 5: Regression Discontinuity Design (this week!)
  - Module 6: Instrumental Variable
  - Module 10: Difference-in-Difference

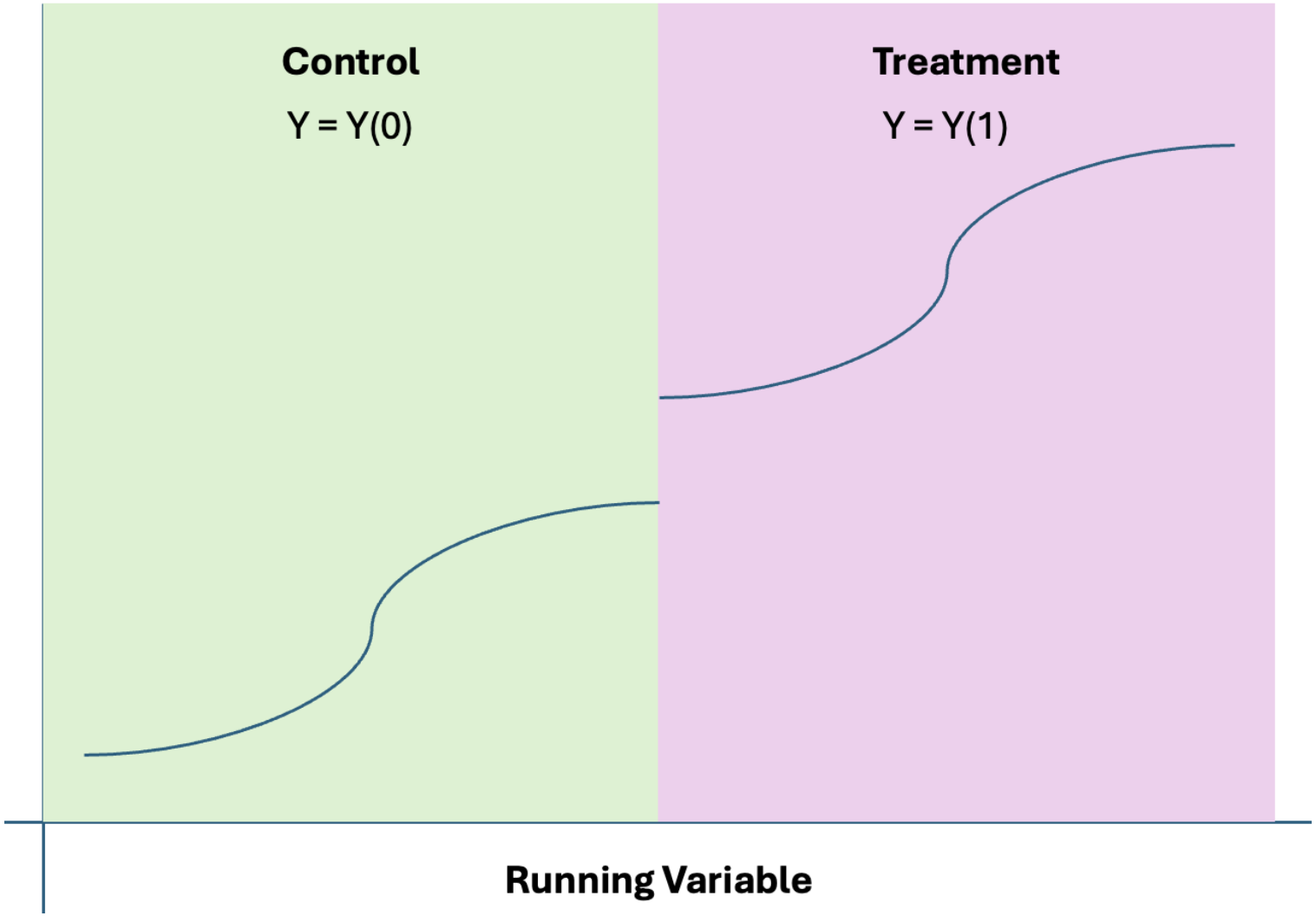
Sharp RDD: Basic Setup and Intuition

- $T_i \in \{0, 1\}$: Treatment
- X_i: **Running variable** that perfectly determines the value of T_i with the cutpoint c

$$T_i = \mathbf{1}\{X_i \geq c\} = \begin{cases} 1 & \text{if } X_i \geq c \\ 0 & \text{if } X_i < c \end{cases}$$

- X_i may be correlated with $Y_i(0)$ and $Y_i(1)$
- Simply adjusting running variable X_i does not work because of lack of overlap assumption
- **Intuition**: At the cutpoint $X_i = c$, assignment to treatment may be as if random
  - Only thing that differs is treatment assignment

Sharp RDD: Basic setup



Sharp RDD: Estimand and Assumption

- **Estimand**: Average treatment effect **on the threshold**

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = c]$$

  - Problem: External validity. **Local** ATE, applicable only to people at the threshold

- **Assumption**: $\mathbb{E}[Y_i(t) \mid X_i = x]$ is continuous in x at $X_i = c$ for $t = 0, 1$
  - Continuity $\rightarrow$ Does not change abruptly
  - Formally, $\lim_{x \to c} \mathbb{E}[Y_i(t) \mid X_i = x] = \lim_{x \leftarrow c} \mathbb{E}[Y_i(t) \mid X_i = x]$
  - Example of violation (sorting): students strategically retaking an exam to just exceed a scholarship cutoff
  - Barely below and above the cutoff is as-if random
    - Not the local randomization (i.e., randomization in the range)

Sharp RDD: More on Continuity Assumption

- In election RD, Grimmer et al. (2011, unpublished) found that winning and losing candidates are very different
  - Candidates and parties anticipate close elections and invest resources strategically.
  - Incumbents and candidates supported by powerful parties may have advantages that affect who wins extremely close contests.
- Question: Does this violate the continuity assumption?

- Not necessarily! The continuity assumption is not local randomization.
  - We need $\lim_{x \to c} \mathbb{E}[Y_i(t) \mid X_i = x] = \lim_{x \leftarrow c} \mathbb{E}[Y_i(t) \mid X_i = x]$, but it does not mean that the covariates are balanced (i.e., randomized)

Sharp RDD: Identification

- Now, the estimand is $\tau = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = c]$
- Then, for $T_i = 1$

$$\begin{aligned}\mathbb{E}[Y_i(1) \mid X_i = c] &= \lim_{x \leftarrow c} \mathbb{E}[Y_i(1) \mid X_i = x] \quad (\because \text{continuity}) \\ &= \lim_{x \leftarrow c} \mathbb{E}[Y_i \mid X_i = x] \quad (\because \text{consistency})\end{aligned}$$

- Similarly, for $T_i = 0$

$$\mathbb{E}[Y_i(0) \mid X_i = c] = \lim_{x \to c} \mathbb{E}[Y_i(0) \mid X_i = x] = \lim_{x \to c} \mathbb{E}[Y_i \mid X_i = x]$$

- Therefore,

$$\tau = \underbrace{\lim_{x \downarrow c} \mathbb{E}[Y_i \mid X_i = x]}_{= \mathbb{E}[Y_i(1) \mid X_i = c]} - \underbrace{\lim_{x \uparrow c} \mathbb{E}[Y_i \mid X_i = x]}_{= \mathbb{E}[Y_i(0) \mid X_i = c]}$$

Sharp RDD: Estimation

- Identification formula is

$$\tau = \underbrace{\lim_{x \downarrow c} \mathbb{E}[Y_i \mid X_i = x]}_{\text{Obtained from above threshold}} - \underbrace{\lim_{x \uparrow c} \mathbb{E}[Y_i \mid X_i = x]}_{\text{Obtained from below Threshold}}$$

- You need to estimate **the cutoff**
- Fit two **local linear regression**: for treated,

$$\arg\min_{\alpha,\beta} \sum_{i=1}^{n} \mathbf{1}\{X_i \geq c\}\{Y_i - \alpha - (X_i - c)\beta\}^2 K\left(\frac{X_i - c}{h}\right)$$

where

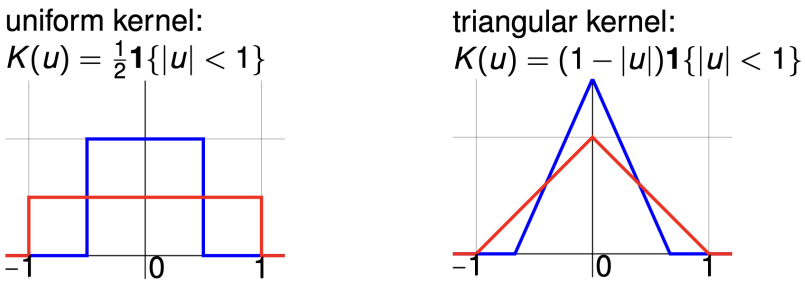
  - $\mathbf{1}\{X_i \geq c\}$: indicator for treated unit (for control, $\mathbf{1}\{X_i \leq c\}$)
  - $K\left(\frac{X_i - c}{h}\right)$ is the weight (kernel)
  - Regressor is centered by $X_i - c$ so that α represnts the intercept at $X_i = c$ (i.e., cutoff)

Local Linear Regression

- **Local Linear Regression**: Fit a linear regression around the cutoff

$$\arg\min_{\alpha,\beta} \sum_{i=1}^{n} \mathbf{1}\{X_i \geq c\} \underbrace{\{Y_i - \alpha - (X_i - c)\beta\}^2}_{\text{Square of Errors}} \underbrace{K\left(\frac{X_i - c}{h}\right)}_{\text{Kernel}}$$

- **Kernel**: Gives more weight around cutoff
  - Recall that we want to model the local behavior around cutoff
  - The regression above is special version of weighted least squares



  - Use triangular kernel (asymptotically optimal; Fan and Gijbels 1996)
- **Bandwidth**: determines how local the regression is
  - Look at the value of running variable $|X_i - c| \leq h$
  - Optimal bandwidth: select bandwidth h so that it minimizes MSE (Imbens and Kalyanaraman 2012)

Review of Weighted Least Square

- **Weighted Least Square**: Weight the observation and solve OLS
- Formally, we minimize

$$\min_{\beta}(Y - X\beta)^\top W(Y - X\beta)$$

- Thus the first order condition is

$$\frac{\partial}{\partial\beta}(Y - X\beta)^\top W(Y - X\beta) = -2X^\top W(Y - X\beta) = 0$$
$$\rightarrow \hat{\beta} = (X^\top WX)^{-1}X^\top WY$$

Local Linear Regression as a Weighted Average (1)

- On the right side of the cutoff, local linear regression solves

$$(\hat{\alpha}_+, \hat{\beta}_+) = \arg\min_{\alpha,\beta} \sum_{i \in \mathcal{I}^+} W_i \{ Y_i - \alpha - \beta \widetilde{X}_i \}^2,$$

where $\widetilde{X}_i = X_i - c$, $\mathcal{I}^+ = \{i : \widetilde{X}_i \geq 0\}$, and $W_i = K(\widetilde{X}_i / h)$.

- The first-order conditions can be written as

$$\underbrace{\begin{bmatrix} \sum_{i \in \mathcal{I}^+} W_i & \sum_{i \in \mathcal{I}^+} W_i \widetilde{X}_i \\ \sum_{i \in \mathcal{I}^+} W_i \widetilde{X}_i & \sum_{i \in \mathcal{I}^+} W_i \widetilde{X}_i^2 \end{bmatrix}}_{\boldsymbol{Z}^\top \boldsymbol{W} \boldsymbol{Z}} \begin{bmatrix} \hat{\alpha}_+ \\ \hat{\beta}_+ \end{bmatrix} = \underbrace{\begin{bmatrix} \sum_{i \in \mathcal{I}^+} W_i Y_i \\ \sum_{i \in \mathcal{I}^+} W_i \widetilde{X}_i Y_i \end{bmatrix}}_{\boldsymbol{Z}^\top \boldsymbol{W} \boldsymbol{Y}}.$$

Local Linear Regression as a Weighted Average (2)

- The inverse of the weighted design matrix is

$$(\boldsymbol{Z}^\top \boldsymbol{W} \boldsymbol{Z})^{-1} = \frac{1}{\left(\sum_{j\in\mathcal{I}^+} W_j\right)\left(\sum_{j\in\mathcal{I}^+} W_j \widetilde{X}_j^2\right) - \left(\sum_{j\in\mathcal{I}^+} W_j \widetilde{X}_j\right)^2} \times \begin{bmatrix} \sum_{j\in\mathcal{I}^+} W_j \widetilde{X}_j^2 & -\sum_{j\in\mathcal{I}^+} W_j \widetilde{X}_j \\ -\sum_{j\in\mathcal{I}^+} W_j \widetilde{X}_j & \sum_{j\in\mathcal{I}^+} W_j \end{bmatrix}.$$

- Taking the first element of $(\boldsymbol{Z}^\top \boldsymbol{W} \boldsymbol{Z})^{-1} \boldsymbol{Z}^\top \boldsymbol{W} \boldsymbol{Y}$ gives

$$\hat{\alpha}_+ = \frac{\left(\sum_{j\in\mathcal{I}^+} W_j \widetilde{X}_j^2\right)\left(\sum_{i\in\mathcal{I}^+} W_i Y_i\right) - \left(\sum_{j\in\mathcal{I}^+} W_j \widetilde{X}_j\right)\left(\sum_{i\in\mathcal{I}^+} W_i \widetilde{X}_i Y_i\right)}{\left(\sum_{j\in\mathcal{I}^+} W_j\right)\left(\sum_{j\in\mathcal{I}^+} W_j \widetilde{X}_j^2\right) - \left(\sum_{j\in\mathcal{I}^+} W_j \widetilde{X}_j\right)^2}.$$

Local Linear Regression as a Weighted Average (3)

- Collecting the terms in each Y_i gives the local-linear weight, which is exactly the expression in the screenshot:

$$\ell_i = \frac{W_i \sum_{j \in \mathcal{I}^+} W_j \widetilde{X}_j^2 - W_i \widetilde{X}_i \sum_{j \in \mathcal{I}^+} W_j \widetilde{X}_j}{\left(\sum_{j \in \mathcal{I}^+} W_j\right)\left(\sum_{j \in \mathcal{I}^+} W_j \widetilde{X}_j^2\right) - \left(\sum_{j \in \mathcal{I}^+} W_j \widetilde{X}_j\right)^2}.$$

- Thus, $\hat{\alpha}_+ = \sum_{i \in \mathcal{I}^+} \ell_i Y_i$ is a weighted sum of the outcomes
- Writing $S_k = \sum_{j \in \mathcal{I}^+} W_j \widetilde{X}_j^k$, this is the compact formula

$$\ell_i = \frac{W_i (S_2 - S_1 \widetilde{X}_i)}{S_0 S_2 - S_1^2}.$$

Key Properties of the Local Linear Weights (1)

- **Property 1**: The equivalent local-linear weights sum to one.

$$\begin{aligned}\sum_{i\in\mathcal{I}^+}\ell_i &= \sum_{i\in\mathcal{I}^+}\frac{W_i(S_2 - S_1\widetilde{X}_i)}{S_0S_2 - S_1^2}\\ &= \frac{S_2\sum_{i\in\mathcal{I}^+}W_i - S_1\sum_{i\in\mathcal{I}^+}W_i\widetilde{X}_i}{S_0S_2 - S_1^2}\\ &= \frac{S_2S_0 - S_1^2}{S_0S_2 - S_1^2} = 1.\end{aligned}$$

Key Properties of the Local Linear Weights (2)

- **Property 2**: The weighted first moment of the centered running variable is zero.

$$\sum_{i\in\mathcal{I}^+} \ell_i \widetilde{X}_i = \sum_{i\in\mathcal{I}^+} \frac{W_i(S_2 - S_1\widetilde{X}_i)\widetilde{X}_i}{S_0 S_2 - S_1^2}$$

$$= \frac{S_2 \sum_{i\in\mathcal{I}^+} W_i\widetilde{X}_i - S_1 \sum_{i\in\mathcal{I}^+} W_i\widetilde{X}_i^2}{S_0 S_2 - S_1^2}$$

$$= \frac{S_2 S_1 - S_1 S_2}{S_0 S_2 - S_1^2} = 0.$$

Why Local Linear Regression is good? (1)

- **Question**: Why local linear regression?
- **Answer**: Local linear regression behaves nicely at the boundary
  - When you estimate a regression near a boundary, bias can appear since there are fewer observations
  - Local linear regression corrects this bias up to the first order
- Let's see why it is the case.
- Recall that we derive the estimator $\hat{\alpha}_+$ is written as

$$\hat{\alpha}_+ = \sum_{i \in \mathcal{I}^+} \ell_i Y_i$$

and thus

$$\mathbb{E}[\hat{\alpha}_+ \mid \tilde{\boldsymbol{X}}] = \sum_{i \in \mathcal{I}^+} \ell_i \underbrace{\mathbb{E}[Y_i \mid \tilde{X}_i]}_{=\mu_1(\tilde{X}_i)}$$

Why Local Linear Regression is good? (2)

- **Goal**: Show $\mathbb{E}[\hat{\alpha}_+ \mid \tilde{\boldsymbol{X}}] = \mathbb{E}[Y_i \mid \tilde{X}_i = 0] + \text{Small Bias}$
- Let's consider Taylor expansion around $\tilde{X}_i = 0$

$$\mu_1(\tilde{X}_i) = \mu_1(0) + \mu_1'(0)\tilde{X}_i + \frac{1}{2}\mu_1''(0)\tilde{X}_i^2 + \cdots$$

- Therefore,

$$\mathbb{E}[\hat{\alpha}_+ \mid \tilde{\boldsymbol{X}}] = \sum_{i \in \mathcal{I}^+} \ell_i \underbrace{\mathbb{E}[Y_i \mid \tilde{X}_i]}_{=\mu_1(\tilde{X}_i)}$$

$$= \sum_{i \in \mathcal{I}^+} \ell_i \left(\mu_1(0) + \mu_1'(0)\tilde{X}_i + \frac{1}{2}\mu_1''(0)\tilde{X}_i^2 + \cdots \right)$$

$$= \mu_1(0) \underbrace{\sum_{i \in \mathcal{I}^+} \ell_i}_{=1} + \mu_1'(0) \underbrace{\sum_{i \in \mathcal{I}^+} \ell_i \tilde{X}_i}_{=0} + \frac{1}{2}\mu_1''(0) \sum_{i \in \mathcal{I}^+} \tilde{X}_i^2 + \cdots$$

Why Local Linear Regression is good? (3)

- **Takeaway**: Bias at the boundary is zero up to first order

$$\mathbb{E}[\hat{\alpha}_+ \mid \tilde{\boldsymbol{X}}] = \mu_1(0) + \underbrace{\frac{1}{2}\mu_1''(0) \sum_{i:\tilde{X}_i \geq 0} \tilde{X}_i^2}_{\text{Bias}} + \cdots$$

  - If you use kth order local polynomial, bias is zero up to k-th order
  - However, as k increases, the variance also increase
- Higher-order bias becomes negligible under optimal bandwidth selection and we can construct valid confidence intervals (Calonico et al. 2014, Econometrica)
  - Recall that we use the observation within bandwidth: i.e., $|\tilde{X}_i| \leq h$
  - As we have more data, we do not need large bandwidth, so $h \to 0$ as $n \to \infty$. So under optimal bandwitdh, $|\tilde{X}_i| \to 0$, which kills the higher-order bias
  - **Takeaway**: Always use the optimal bandwidth

Sharp RDD: Covariate Adjustment

- Covariates that predict Y_i near the cutoff can improve precision
  - But it does not fix the problem if the continuity assumption fails
  - Covariates fix RDD under the stringent conditions (Frölich and Huber 2019)
- Suppose that you add a covariate C_i to the local linear regression

$$Y_i = \alpha + \tau T_i + \beta \tilde{X}_i + \gamma T_i \tilde{X}_i + C_i^\top \boldsymbol{\delta} + \epsilon_i.$$

where C_i is continuous at c: i.e.,

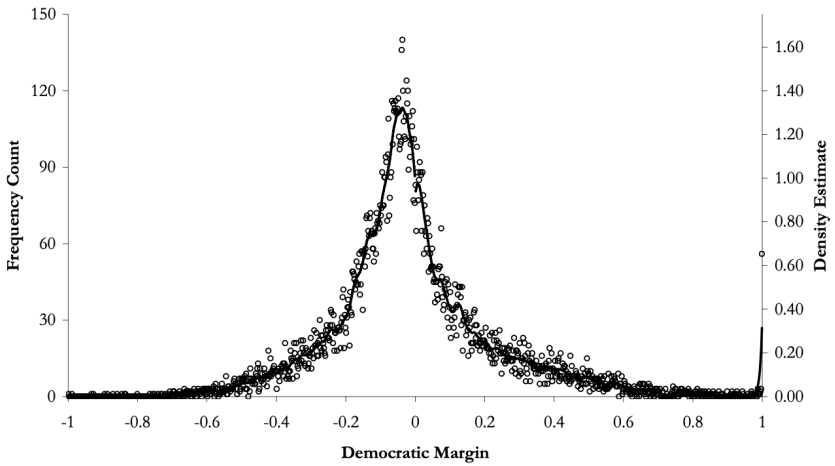
$$\lim_{x \downarrow c} \mathbb{E}[C_i \mid X_i = x] = \lim_{x \uparrow c} \mathbb{E}[C_i \mid X_i = x].$$

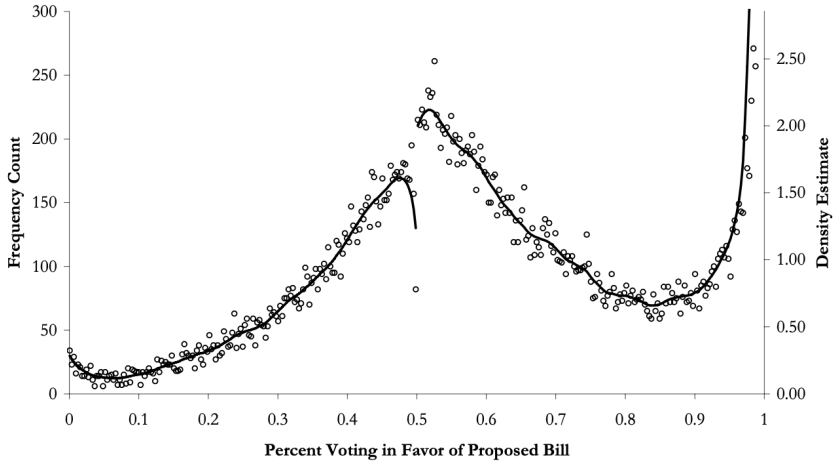
- However, this does not change the estimand τ because the covariate contribution cancels at the cutoff

$$\begin{aligned}
&\lim_{x \downarrow c} \mathbb{E}[Y_i - C_i^\top \boldsymbol{\delta} \mid X_i = x] - \lim_{x \uparrow c} \mathbb{E}[Y_i - C_i^\top \boldsymbol{\delta} \mid X_i = x] \\
&= \lim_{x \downarrow c} \mathbb{E}[Y_i \mid X_i = x] - \lim_{x \uparrow c} \mathbb{E}[Y_i \mid X_i = x] \\
&\quad - \underbrace{\left\{ \lim_{x \downarrow c} \mathbb{E}[C_i \mid X_i = x] - \lim_{x \uparrow c} \mathbb{E}[C_i \mid X_i = x] \right\}^\top}_{=0} \boldsymbol{\delta}
\end{aligned}$$

Diagnosis on Sorting

- McCrary Test: Test for sorting by checking the continuity of the density of the running variable at the cutoff
  - If treatment assignment is manipulated, the density of the running variable will be discontinuous at the cutoff





- Another approach: Checking continuity of covariates

- Both are indirect test of continuity in $\mathbb{E}[Y_i(t) \mid X_i = x]$
  - You should test both if possible

When continuity assumption fails (1)

- Suppose that the continuity assumption fails because of misreporting of the running variable X_i on one side of the cutoff
  - For example, households may misreport their income to be eligible for a program

- **Setup**
  - X_i: self-reported income
  - $T_i = \mathbf{1}\{X_i \geq c\}$: treatment indicator
    - whether the household is eligible for the program (c is threshold)
  - Y_i: outcome of interest (continuous)
  - M_i: misreporting status

- **Assumption**:
  - $\mathbb{E}[Y_i(t) \mid X_i = x, M_i = 0]$ is continuous
    - I.e., among those who do not misreport, continuity holds
  - If $X_i < c$, then $M_i = 0$
    - No units with $X_i < c$ are manipulators

When continuity assumption fails (2)

- Let $f_{X,M}(x, m)$ be the joint density of $X = x, M = m$.
- Above the cutoff, we have mixture of manipulators and non-manipulators

$$f_+(c) := \lim_{x \uparrow c} f_X(x) = f_{X,M}(c, 0) + f_{X,M}(c, 1)$$

while below the cutoff we only have the nonmanpulators

$$f_-(c) := \lim_{x \downarrow c} f_X(x) = f_{X,M}(c, 0)$$

- Therefore,

$$\mathbb{P}(M_i = 1 \mid X_i = c) = \frac{f_{X,C}(c, 1)}{f_+(c)} = \frac{f_+(c) - f_-(c)}{f_+(c)}$$

Extra: When continuity assumption fails (3)

- Now, let's think about the bounds of LATE

$$\mathbb{E}\big[Y_i(1) - Y_i(0) \mid X_i = c, M_i = 0\big]$$

- As everyone below the cutoff is non-manipulator, by continuity

$$\mathbb{E}\big[Y_i(0) \mid X_i = c, M_i = 0\big] = \lim_{x \uparrow c} \mathbb{E}[Y_i \mid X_i = x]$$

  which is point identified.

- We thus need to bound

$$\mathbb{E}\big[Y_i(1) \mid X_i = c, M_i = 0\big]$$

  since just above the cutoff, we have mixture of manipulators and nonmanipulators

Extra: When continuity assumption fails (4)

- However, now we know how many people are manipulators at the cutoff; i.e.,

$$p = \mathbb{P}(M_i = 1 \mid X_i = c) = \frac{f_+(c) - f_-(c)}{f_+(c)}$$

  which means that $1 - p$ people are non-manipulators
- We also observe the distribution of outcomes
- **Idea**: Think about how to allocate these $1 - p$ people
  - If everyone is at the bottom of outcome distribution, we then obtain the lower bounds
  - If everyone is at the top of outcome distribtuion, we then obtain the upper bounds
  - Formally, with quantile function $Q^+(u) = \inf\{y : F^+(y) \geq u\}$

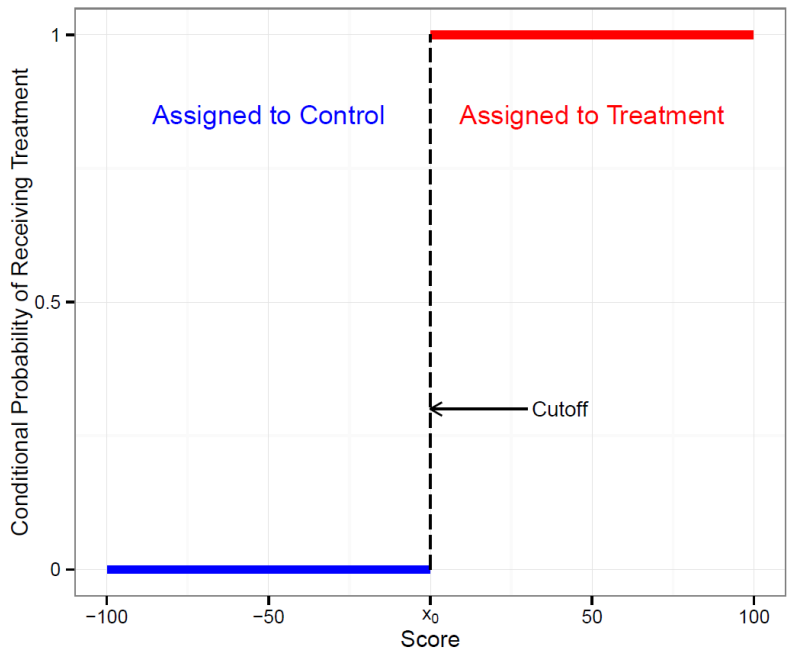
$$\underline{\mu}_1 = \frac{1}{1-p} \int_0^{1-p} Q^+(u)\, du, \qquad \overline{\mu}_1 = \frac{1}{1-p} \int_p^1 Q^+(u)\, du$$

- This is called Lee's bounds

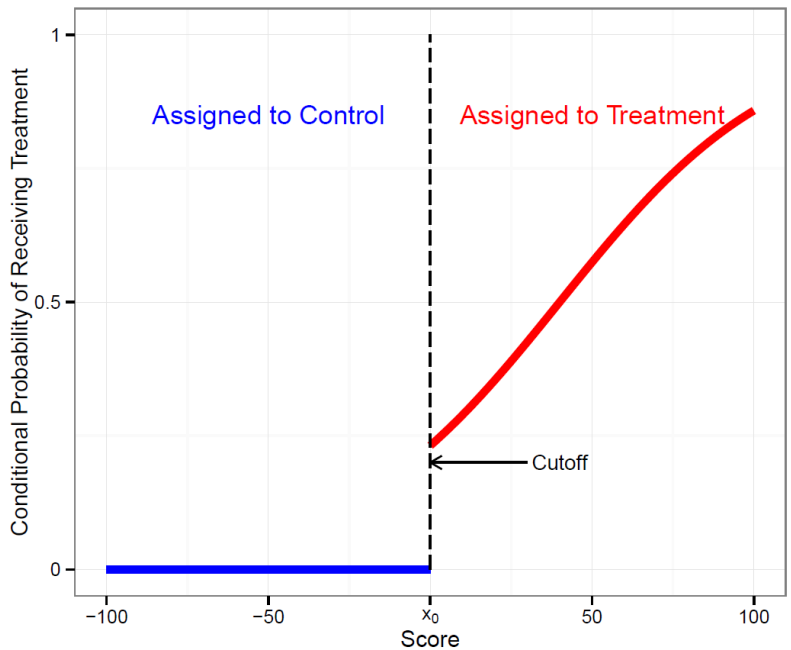
Fuzzy RDD: Intuition

- **Fuzzy RDD**: Instrumental variable version of RDD
  - You have a running variable that does not perfectly determine the treatment status
  - e.g., eligibility of enrollment

Figure 1: Conditional Probability of Receiving Treatment in Sharp vs. Fuzzy RD Designs



(a) Sharp RD



(b) Fuzzy RD (One-Sided)

Fuzzy RDD: Setup and Estimand

- Setup
  - $Z_i = \mathbf{1}\{X_i \geq c\}$: instrument (e.g., eligibility for a program)
    - Instrument is determined by running variable
  - T_i: treatment (e.g., actual enrollment in a program)
  - Y_i: outcome
  - Potential outcome: $Y_i(Z_i = z, T_i = t)$

- **Estimand**: LATE among complier at threshold

$$\mathbb{E}[Y_i(Z_i = 1, T_i = T_i(1)) - Y_i(Z_i = 0, T_i = T_i(0)) \mid X_i = c, \text{Complier}]$$

Fuzzy RDD: Assumption and Identification

- **Assumptions**: IV + Sharp RDD
  - Monotonicity: $T_i(1) \geq T_i(0)$
  - Exclusion restriction: $Y_i(0, t) = Y_i(1, t)$
  - Continuity: $\mathbb{E}[T_i(z) \mid X_i = x]$ and $\mathbb{E}[Y_i(z, T_i(z)) \mid X_i = x]$ are continuous in x

- Recall that the identification formula of IV is given by

$$\frac{\text{Instrument Effect on Outcome (Z's effect on Y)}}{\text{Instrument Effect on Treatment (Z's effect on T)}}$$

- As Instrument's local effect is estimated by RDD, we use

$$\frac{\lim_{x \downarrow c} \mathbb{E}[Y_i \mid X_i = x] - \lim_{x \uparrow c} \mathbb{E}[Y_i \mid X_i = x]}{\lim_{x \downarrow c} \mathbb{E}[T_i \mid X_i = x] - \lim_{x \uparrow c} \mathbb{E}[T_i \mid X_i = x]}$$

Review Question: Setting

- **Question**: Do politicians get richer by serving in Parliament?
  - Eggers and Hainmueller (2009): estates of deceased British candidates who ran in elections between 1950 and 1970
  - Winning is not randomly assigned: winners and losers may differ in ways that also affect wealth

- Data: Tory candidates (`mpforsale_tory.csv`), $n = 222$

	Variable	Description
Y_i	`lnrealgross`	Log of gross wealth at death
T_i	`treated`	Indicator for winning the election
X_i	`margin`	Vote share margin (running variable)
	`pre_margin`	Previous election's vote share margin

- $T_i = \mathbf{1}\{X_i \geq 0\} \rightarrow$ Sharp RDD with cutoff $c = 0$

Review Question 1

- Use candidates within ± 2 percentage points of the cutoff (i.e., $|X_i| \leq 0.02$)
  - Scatter plot of Y_i against X_i with a dashed vertical line at the cutoff
  - Fit two separate linear regressions (winners and losers) and plot the fitted lines with 95% CIs
- Report the estimated ATE at the cutoff with the heteroskedasticity robust standard error and 95% CI
  - Tip: `lm_robust()` in the `estimatr` package
- Give a substantive interpretation of the plot and estimate
- Discuss the assumption required for identification at the cutoff based on model extrapolation. When might it be violated?

Review Question 1 - Answer (1)

- Two separate regressions within the window $|X_i| \leq h$:

$$\text{Losers } (X_i < 0): \quad Y_i = \alpha_0 + \beta_0 X_i + \epsilon_i$$
$$\text{Winners } (X_i \geq 0): \quad Y_i = \alpha_1 + \beta_1 X_i + \epsilon_i$$

  Since $c = 0$, the intercepts α_0, α_1 are the predicted outcomes at the cutoff

- Equivalently, fit one regression with an interaction:

$$Y_i = \alpha + \tau T_i + \beta X_i + \gamma T_i X_i + \epsilon_i$$

  - Losers: intercept α, slope β
  - Winners: intercept $\alpha + \tau$, slope $\beta + \gamma$
  - Thus, $\hat{\tau} = \hat{\alpha}_1 - \hat{\alpha}_0$ is the jump at the cutoff

- This is the local linear regression with a uniform kernel $K(u) = \mathbf{1}\{|u| \leq 1\}$ and bandwidth $h = 0.02$

Review Question 1 - Answer (2)

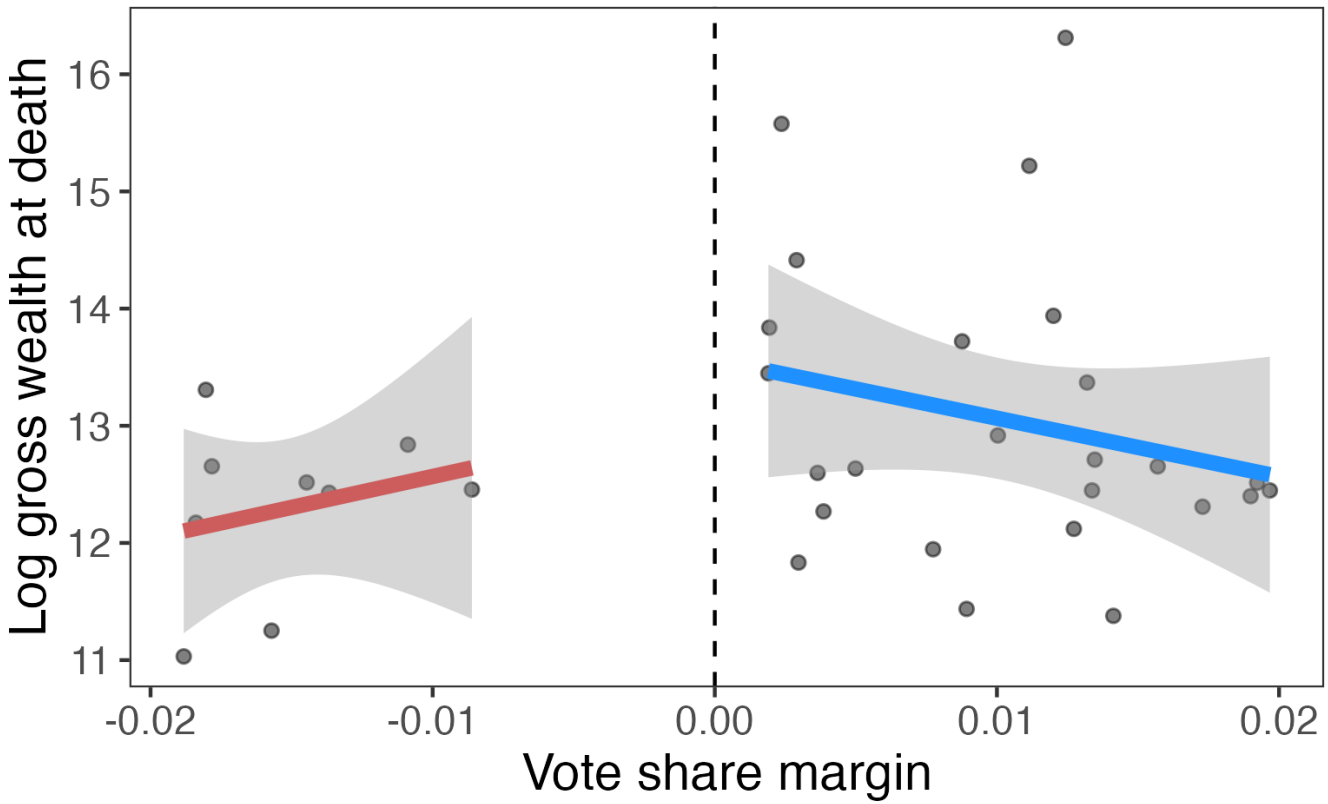
- Plot the two fitted lines within the window

```
library(tidyverse)
data <- read_csv("mpforsale_tory.csv")

bw_q1 <- 0.02
data_q1 <- data %>% filter(abs(margin) <= bw_q1)

data_q1 %>%
  ggplot(aes(x = margin, y = lnrealgross)) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  geom_point(alpha = 0.5) +
  geom_smooth(data = data_q1 %>% filter(margin < 0),  # losers
              method = "lm", se = TRUE,
              color = "IndianRed", linewidth = 2) +
  geom_smooth(data = data_q1 %>% filter(margin >= 0), # winners
              method = "lm", se = TRUE,
              color = "DodgerBlue", linewidth = 2) +
  labs(x = "Vote share margin", y = "Log gross wealth at death")
```

Review Question 1 - Answer (3)



- Visual jump at the cutoff: barely-winning candidates seem wealthier at death than barely-losing candidates
- But only 34 candidates (9 losers, 25 winners) $\rightarrow$ wide confidence bands

Review Question 1 - Answer (4)

- Fit the interacted model within the window (`lm_robust()` uses HC2 by default)

```
library(estimatr)
rdd_lm <- lm_robust(lnrealgross ~ treated * margin, data = data,
                    subset = abs(margin) <= bw_q1)
```

Term	Estimate	Std. Error	CI Lower	CI Upper
(Intercept)	13.095	0.789	11.484	14.707
`treated`	0.468	0.920	−1.410	2.346
`margin`	52.784	62.957	−75.793	181.360
`treated:margin`	−102.617	71.536	−248.712	43.478

Review Question 1 - Answer (5)

- $\hat{\tau} = 0.468$ (robust SE $= 0.920$): ATE **at the cutoff**
  - Outcome is in logs: $e^{0.468} - 1 \approx 0.60$
  - Barely winning a seat raises wealth at death by about 60% for candidates in close elections

- 95% CI: $[-1.410, 2.346]$ covers zero
  - Cannot reject the null of no effect at the 5% level

- Still, the point estimate is substantively large
  - The CI is wide because only 34 candidates are in the window
  - Failing to reject $\neq$ no effect

Review Question 1 - Answer (6)

- No one is observed under both treatments at $X_i = 0$: $\hat{\tau} = \hat{\alpha}_1 - \hat{\alpha}_0$ **extrapolates** each fitted line to the cutoff
- **Assumption**: $\mathbb{E}[Y_i(t) \mid X_i = x]$ is continuous in x at $x = 0$ for $t = 0, 1$
  - Also, the linear fit should be good within the window
- **Violation**: sorting around the cutoff
  - e.g., wealthy candidates "buy" close elections $\rightarrow$ barely-winners are richer even without office
- **Checks** (observable implications)
  - McCrary density test: no jump in the density of X_i at 0
  - Placebo test: no jump in pre-treatment covariates (e.g., `pre_margin`)

Review Question 2

- How sensitive is the estimate in Question 1 to the choice of bandwidth?
- Re-estimate the model for each bandwidth $h \in \{0.02, 0.03, \ldots, 0.10\}$ (i.e., 2 to 10 percentage points)
  - Plot the estimates with 95% CIs
- Add the estimate based on the **MSE-optimal bandwidth**
  - Use a single bandwidth for both sides of the cutoff
  - Tip: `rdbwselect()` in the `rdrobust` package
- Give a substantive interpretation of the plot

Review Question 2 - Answer (1)

- Bandwidth h controls a **bias-variance tradeoff**
  - Small h: linear approximation is good near the cutoff (small bias), but few observations (large variance)
  - Large h: many observations (small variance), but units far from the cutoff may follow a different trend (large bias)

- **MSE-optimal bandwidth**: choose h to minimize

$$\text{MSE}(h) = \text{Bias}(h)^2 + \text{Variance}(h)$$

  - Recall: bias of local linear regression comes from the second-order term $\mu''(0)$
  - `bwselect = "mserd"`: one common MSE-optimal bandwidth for both sides

Review Question 2 - Answer (2)

- Turn Question 1 into a function of the bandwidth

```
bandwidth <- seq(from = 0.02, to = 0.1, by = 0.01)

# Estimate the jump at the cutoff for a given bandwidth
RD_estimator <- function(bw) {
  rdd_lm <- lm_robust(lnrealgross ~ treated * margin,
                      data = data, subset = abs(margin) <= bw)
  rdd_lm %>% tidy() %>%
    filter(term == "treated") %>%
    select(estimate, conf.low, conf.high) %>%
    mutate(bw = bw, colour = "Sequence")
}

# Apply to each bandwidth
estimates <- map_df(bandwidth, RD_estimator)
```

Review Question 2 - Answer (3)

- Compute the MSE-optimal bandwidth and fit the same model

```
library(rdrobust)
opt_bw <- rdbwselect(y = data$lnrealgross,
                     x = data$margin,
                     c = 0,                    # cutoff
                     kernel = "triangular",  # default
                     bwselect = "mserd")$bws[1]
opt_bw # 0.0619

opt_bw_lm <- lm_robust(lnrealgross ~ treated * margin,
                       data = data, subset = abs(margin) <= opt_bw)
opt_estimate <- opt_bw_lm %>% tidy() %>%
  filter(term == "treated") %>%
  select(estimate, conf.low, conf.high) %>%
  mutate(bw = opt_bw, colour = "Optimal Bandwidth")
```

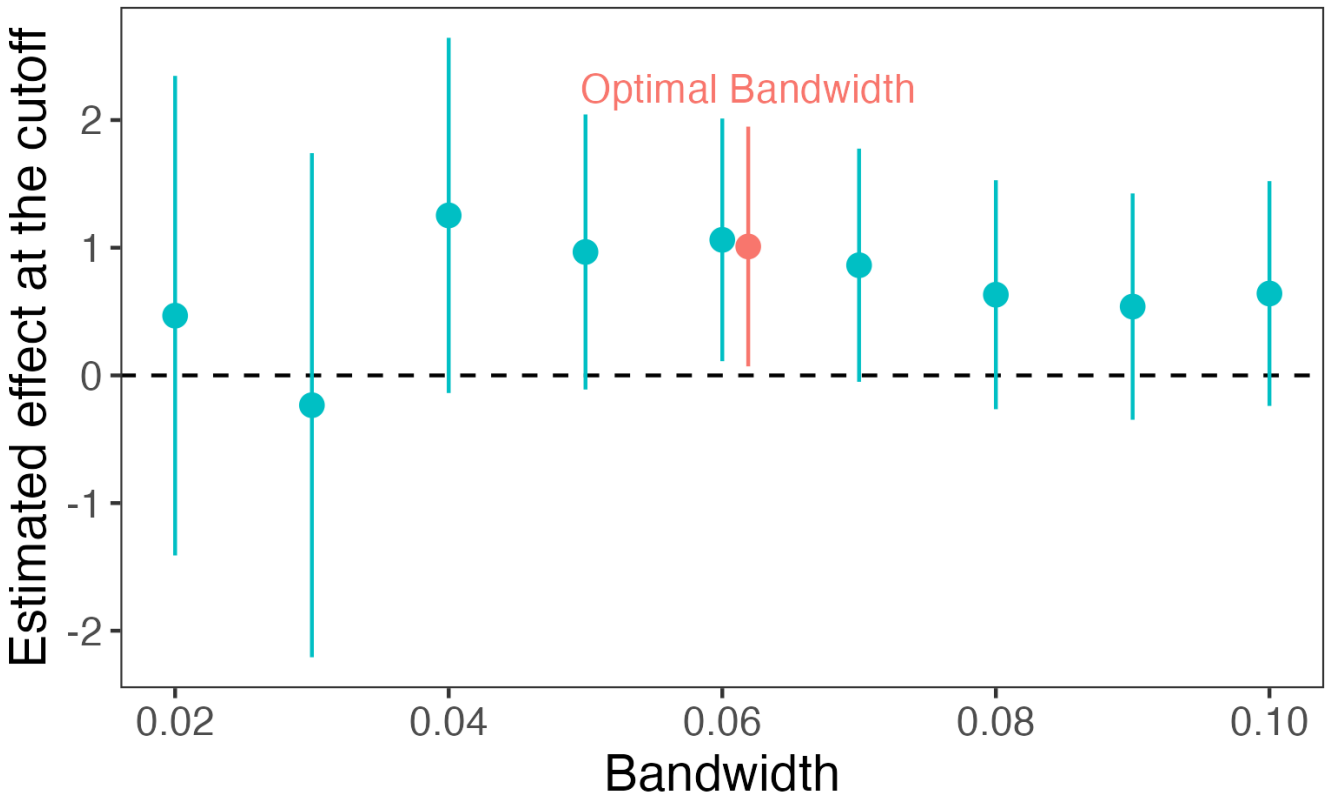
- $h^* \approx 0.062$: about 6 percentage points on each side (90 candidates)

Review Question 2 - Answer (4)

- Plot the estimates against the bandwidth

```
rbind(estimates, opt_estimate) %>%
  ggplot(aes(x = bw, y = estimate,
             ymin = conf.low, ymax = conf.high)) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_pointrange(aes(colour = colour), show.legend = FALSE) +
  annotate("text", x = opt_bw,
           y = opt_estimate$conf.high + 0.3,
           label = "Optimal Bandwidth", colour = "#F8766D") +
  labs(x = "Bandwidth", y = "Estimated effect at the cutoff")
```

Review Question 2 - Answer (5)



- With $h^* \approx 0.062$: $\hat{\tau} = 1.010$, 95% CI $[0.071, 1.949] \rightarrow$ significant at the 5% level
- But the estimates range from -0.23 to 1.25 across bandwidths

Review Question 2 - Answer (6)

Bandwidth	n	Estimate	Std. Error	95% CI
0.02	34	0.468	0.920	$[-1.410, 2.346]$
0.03	50	-0.233	0.981	$[-2.208, 1.741]$
0.04	60	1.253	0.694	$[-0.138, 2.644]$
0.05	73	0.967	0.540	$[-0.111, 2.044]$
0.06	89	1.062	0.478	$[0.111, 2.012]$
$h^* = 0.062$	90	1.010	0.472	$[0.071, 1.949]$
0.07	98	0.863	0.460	$[-0.050, 1.776]$
0.08	108	0.632	0.452	$[-0.265, 1.529]$
0.09	121	0.539	0.448	$[-0.347, 1.425]$
0.10	129	0.641	0.445	$[-0.239, 1.522]$

- As h grows, n increases and the standard error shrinks
- CIs exclude zero only around $h = 0.06$

Review Question 2 - Answer (7)

- At the MSE-optimal bandwidth, barely winning a seat raises wealth at death by $e^{1.010} - 1 \approx 1.75$ (i.e., about 175%)
  - Evidence of a large financial return to office among Tories
- However, the result is **sensitive to the bandwidth**
  - Small h: very wide CIs (few observations)
  - Large h: narrower CIs, but estimates shrink to about 0.6 (possibly bias from units far from the cutoff)
- **Takeaway**: Always report the sensitivity to the bandwidth
  - Do not pick the bandwidth that gives a significant result

Review Question 2: Remark (Extra)

- Two caveats on the analysis above
  - h^* is optimal for the **triangular** kernel, but `lm_robust()` on a subset uses the **uniform** kernel
  - The conventional CI ignores the bias from the linear approximation. With the MSE-optimal h^*, this bias is not negligible relative to the standard error $\rightarrow$ undercoverage

- `rdrobust()` fixes both: triangular kernel + **robust bias-corrected** CI (Calonico, Cattaneo, and Titiunik 2014)

```
rdrobust(y = data$lnrealgross, x = data$margin, c = 0)
```

- Results with $h^* \approx 0.062$
  - Conventional: $\hat{\tau} = 1.160$ (SE $= 0.438$)
  - Robust bias-corrected 95% CI: $[0.235, 2.384]$

Extra: Higher-order bias correction (Intuition)

- Recall that we showed that

$$\mathbb{E}[\hat{\alpha}_+ \mid \tilde{\boldsymbol{X}}] = \mu_1(0) + \underbrace{\frac{1}{2}\mu_1''(0) \sum_{i:\tilde{X}_i \geq 0} \tilde{X}_i^2 + \cdots}_{\text{Bias}}$$

  which shows that we do have the second-order bias
- Recall that we use the observation within bandwidth: i.e., $|\tilde{X}_i| \leq h$
  - If we use the optimal bandwidth, we can show that $h = O(n^{-1/5})$ (i.e., the bandwidth becomes more narrow as the sample size increases)
  - Using this, we can show that the third-order bias is negligible under optimal bandwidth selection
- The second-order bias is not negligible for variance / standard error estimation, so typically we estimate the second-orderbias (by calculating $\mu_1''(0)$) and correct the bias (Calonico et al. 2014, Econometrica)
  - Including the second-order term in local polynomial regression changes the order of the optimal bandwidth, which does not necessarily make the third-order bias negligible

Extra: Higher-order bias correction: Formal (1)

- This is for the sake of completeness, but we will not go into the details of this part
- The optimal bandwidth minimizes the $\mathrm{MSE} = \mathrm{Bias}^2 + \mathrm{Variance}$
  - We already show that the bias is second order (i.e., $\mathrm{Bias}^2 = O(h^4)$)
  - This part will change if we include the second-order term in local polynomial regression
- On the other hand, variance is approximately written as

$$\frac{\text{Variance (constant and does not change by sample size)}}{\text{Effective Sample Size}},$$

  so we want to see what is the order of the effective sample size.
  - Now, approximately $\Pr(c \leq X_i \leq c + h) \approx f_X(c)h$
  - So, Effective Sample Size $\approx n \Pr(c \leq X_i \leq c + h) \approx n f_X(c)h$
- As a result, with some constant C_B and C_V, we can write the MSE as

$$\mathrm{MSE} = C_B h^4 + \frac{C_V}{n f_X(c) h} \implies h^* = O(n^{-1/5})$$

Extra: Higher-order bias correction: Formal (2)

- Now, notice that

$$\frac{\hat{\tau} - \tau}{\sqrt{\text{Var}(\hat{\tau})}} = \frac{\hat{\tau} - \mathbb{E}[\hat{\tau}]}{\sqrt{\text{Var}(\hat{\tau})}} + \frac{\mathbb{E}[\hat{\tau}] - \tau}{\sqrt{\text{Var}(\hat{\tau})}}$$

- The first term is asymptotically normal by CLT, but the second term is not negligible
  - The numerator is bias, which is $O(h^2) = O(n^{-2/5})$
  - The denominator is standard error, which is $O(1/\sqrt{nh}) = O(n^{-2/5})$
  - As a result, the reminder term does not vanish even for the large sample size
- The leading bias of the local-linear estimator depends on the curvature of the conditional mean:

$$\mathbb{E}[\hat{\tau}] - \tau = Bh^2 + O(h^3),$$

where B depends on the second derivatives of the conditional mean functions on the two sides of the cutoff.

Extra: Higher-order bias correction: Formal (3)

- We can estimate B using a local quadratic regression and define the bias-corrected estimator

$$\hat{\tau}_{\mathrm{bc}} = \hat{\tau} - h^2 \hat{B}.$$

- Substituting the bias expansion,

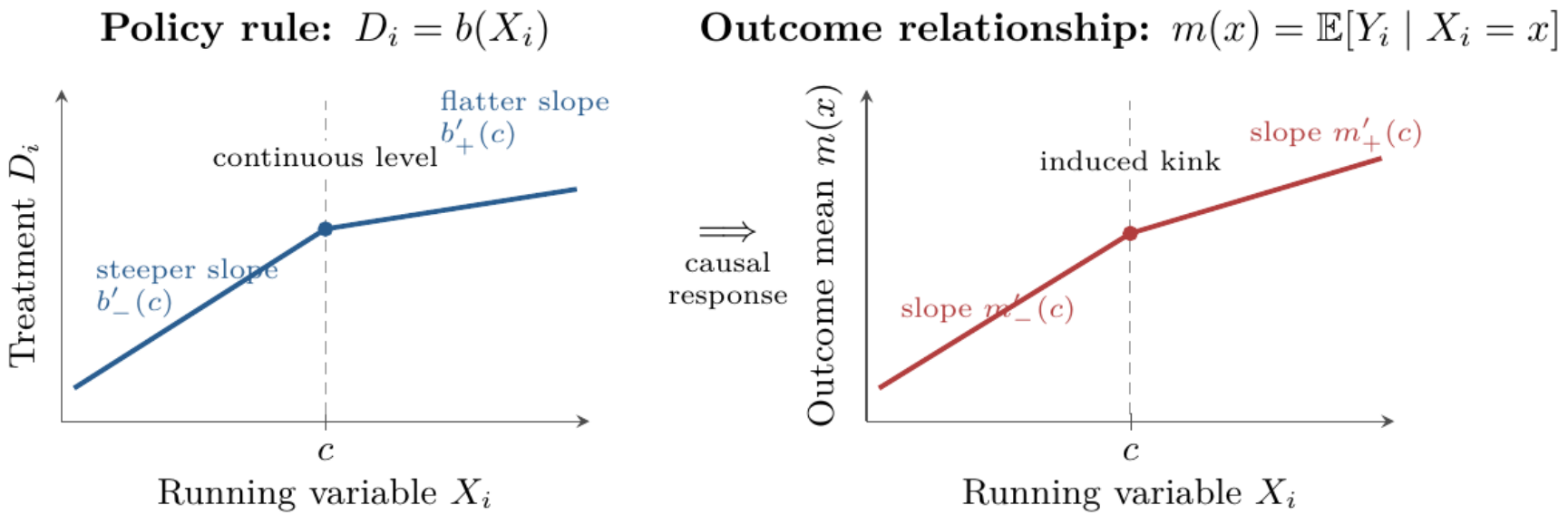
$$\begin{aligned} \hat{\tau}_{\mathrm{bc}} - \tau &= \hat{\tau} - \tau - h^2 \hat{B} \\ &= \{\hat{\tau} - \mathbb{E}[\hat{\tau}]\} - h^2(\hat{B} - B) + O(h^3). \end{aligned}$$

  and the last error $O(h^3)$ is negligible relative to the standard error of $\hat{\tau}_{\mathrm{bc}}$.
- However, because $\hat{B}$ is estimated, robust bias-corrected inference must account for the additional variation from $h^2 \hat{B}$.
  - You need to consider the covariance between $\hat{\tau}$ and $\hat{B}$ to get the correct standard error. I.e.,

$$\mathrm{Var}(\hat{\tau}_{\mathrm{bc}}) = \underbrace{\mathrm{Var}(\hat{\tau})}_{\text{Original}} + h^4 \mathrm{Var}(\hat{B}) - 2h^2 \mathrm{Cov}(\hat{\tau}, \hat{B})$$

Extra: Regression Kink Design (RKD) (1)

- In RDD, the level of treatment jumps at the cutoff; in RKD, treatment remains continuous but its **slope** changes
  - Example: benefits rise with previous earnings until a cap, then flatten



- **Intuition**: If treatment causes the outcome, a kink in the policy rule should induce a kink in the outcome relationship at the same cutoff
  - We cannot use fuzzy RD here since treatment assignment is continuous at the cutoff

Extra: Regression Kink Design (RKD) (2)

- **Setup**: A continuous treatment is assigned by a policy rule

$$D_i = b(X_i), \qquad b(c^-) = b(c^+), \qquad \underbrace{b'_-(c) \neq b'_+(c)}_{\text{kink at } c}.$$

- **Key identifying assumption**: In the absence of the policy kink, the outcome relationship would be smooth in slope at c
  - In the additive model $Y_i = \tau D_i + g(X_i) + \varepsilon_i$, require $g'(x)$ and $\frac{d}{dx}\mathbb{E}[\varepsilon_i \mid X_i = x]$ to be continuous at c
  - Thus, no other policy, behavioral response, or compositional change may create a kink at exactly the same cutoff
- **Estimand**: The local marginal causal effect of treatment at the cutoff

$$\tau_{\text{RKD}} = \frac{\partial}{\partial d}\mathbb{E}[Y_i(d) \mid X_i = c]\Big|_{d=b(c)}$$

Extra: Regression Kink Design (RKD) (3)

- Let $m(x) = \mathbb{E}[Y_i \mid X_i = x]$ and $r(x) = g(x) + \mathbb{E}[\varepsilon_i \mid X_i = x]$. Under the additive model,

$$m'(x) = \tau b'(x) + r'(x).$$

- Taking the right derivative minus the left derivative at c gives

$$m'_+(c) - m'_-(c) = \tau\{b'_+(c) - b'_-(c)\} + \underbrace{\{r'_+(c) - r'_-(c)\}}_{=0 \text{ by smoothness}}$$

$$= \tau\{b'_+(c) - b'_-(c)\}.$$

- Therefore,

$$\tau_{\mathrm{RKD}} = \frac{m'_+(c) - m'_-(c)}{b'_+(c) - b'_-(c)} = \frac{\text{kink in the outcome}}{\text{kink in the treatment}}.$$

- **Estimation**: Fit local polynomial regressions on each side to estimate the one-sided derivatives.